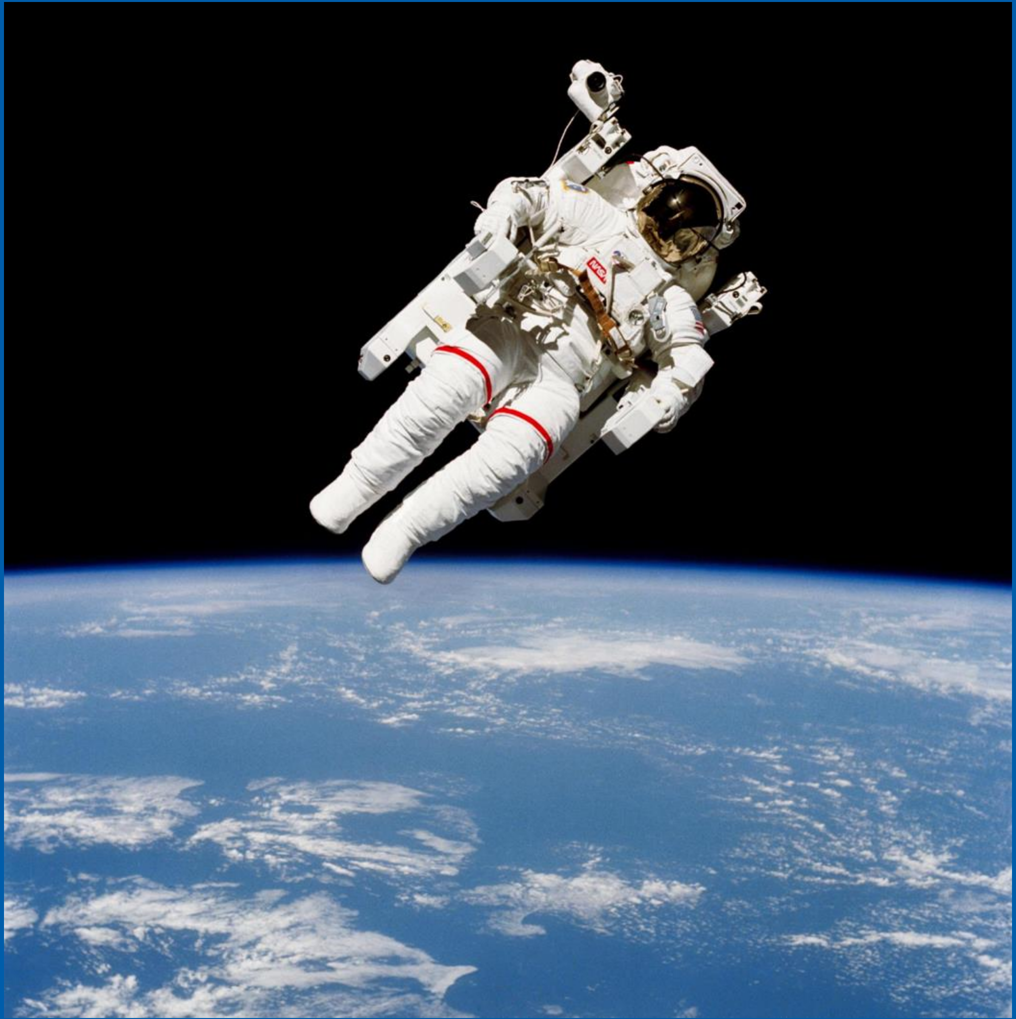
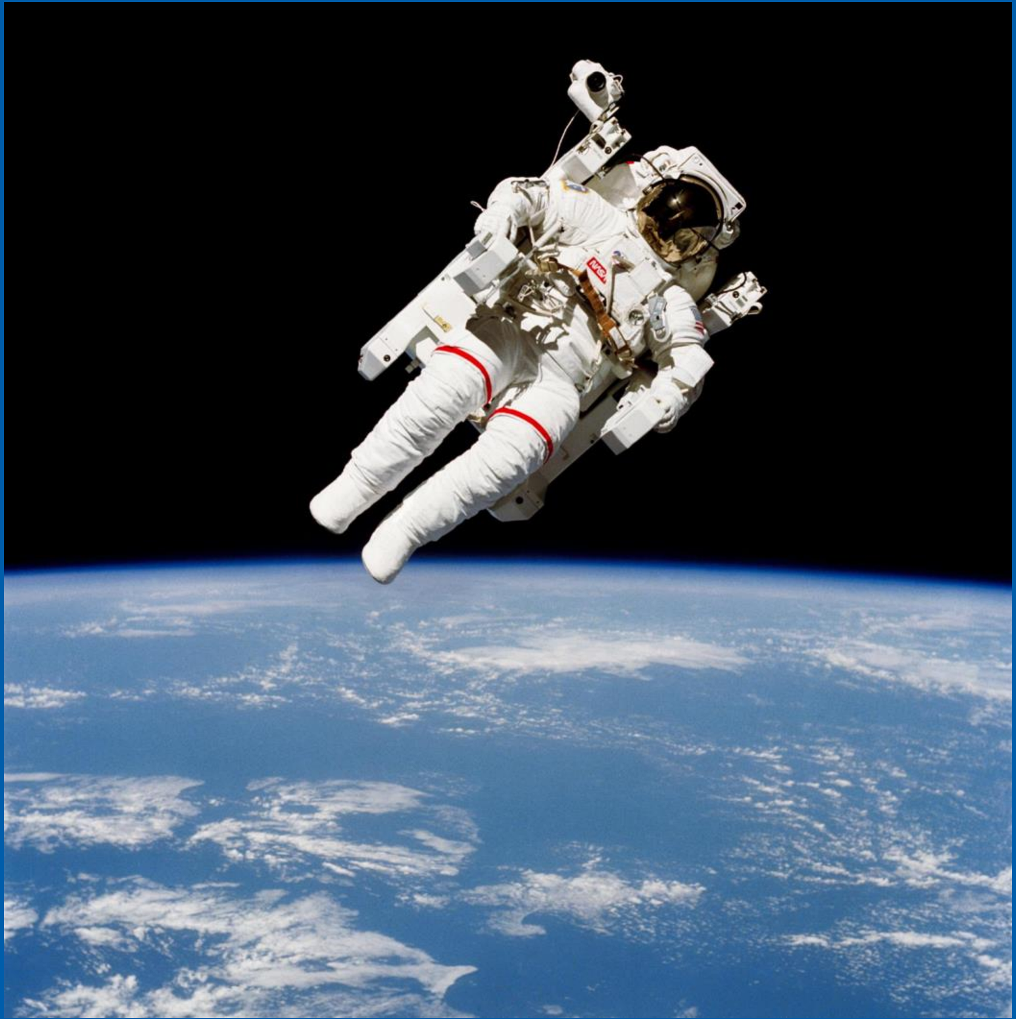
CUNY GRADUATE CENTER

Astronomy 102 - Lab 04 - Motion & Gravity

Matthew Prest



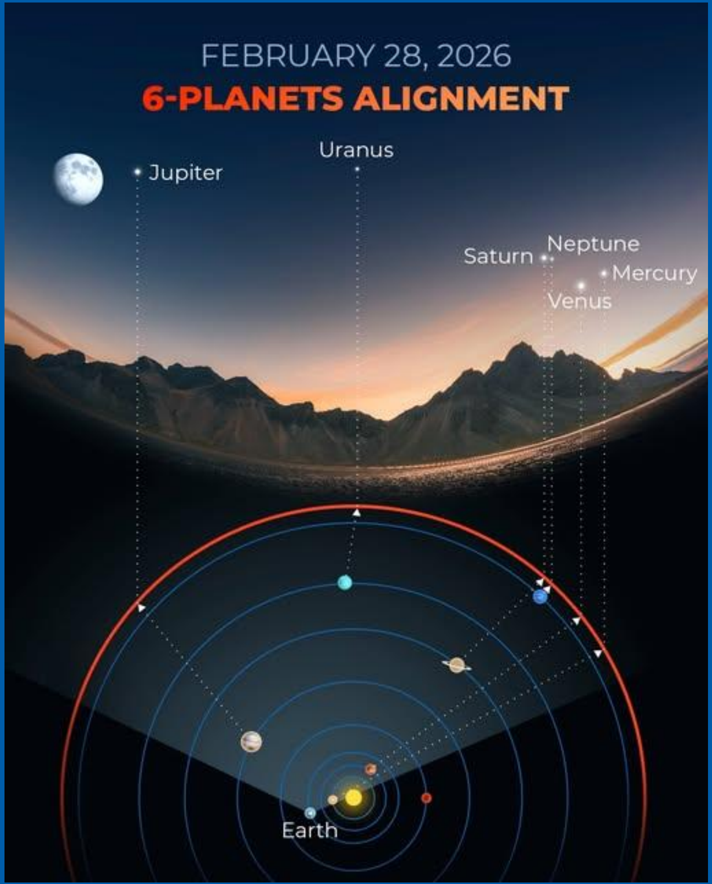
Astronaut Bruce McCandless II flew the Manned Maneuvering Unit out of the space shuttle Challenger's payload bay for the first time on February 7, 1984.



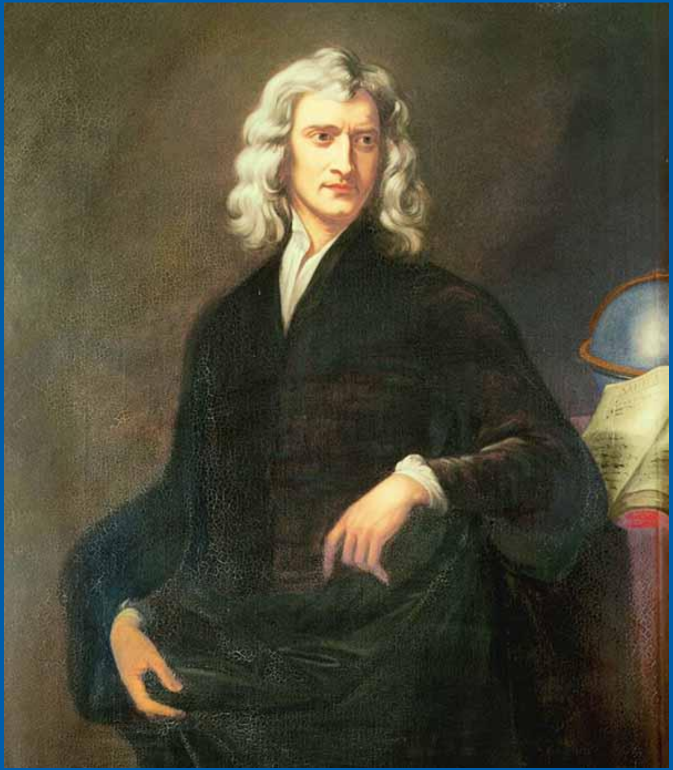

The latest in Astronomy

Six Planet Alignment

- Visible Saturday Feb 28th
- Mercury, Venus, Jupiter, Saturn, Uranus and Neptune will all be visible at the same time
- The best time is 6:16pm
- Most of the easily-visible planets – Saturn, Venus, Mercury – will be near the horizon to the west, with Jupiter higher in the sky and more east.
- Saturn, Venus and Jupiter are very bright, and Saturn is a distinctive yellow colour.
- Mercury can be seen with the naked eye but will be close to the horizon and can be tricky to spot. It will be a similar height and just to the right of Venus.
- Neptune and Uranus require binoculars or a telescope



Earthrise, taken on December 24, 1968, by Apollo 8 astronaut William Anders



If I have seen further it is by standing on the shoulders of Giants."
— Isaac Newton, 1643-1727

"I can calculate the motion of heavenly bodies but not the madness of people."
— Isaac Newton

Key Concepts

- What’s the difference between speed and velocity?

- What’s the difference velocity and acceleration?

Key Concepts

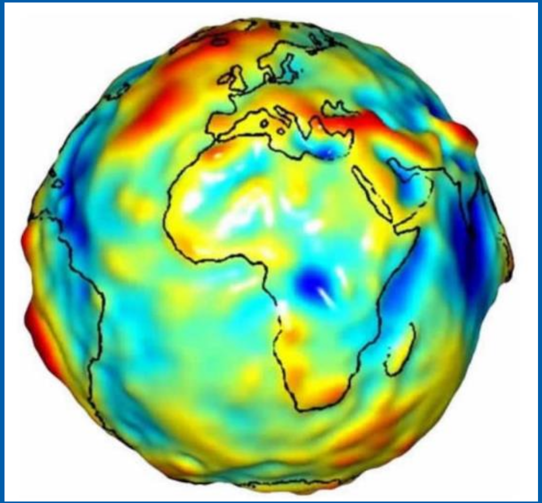
- **What's the difference between speed and velocity?**
  - Velocity has a direction! It can be positive or negative, speed is the "magnitude" of velocity

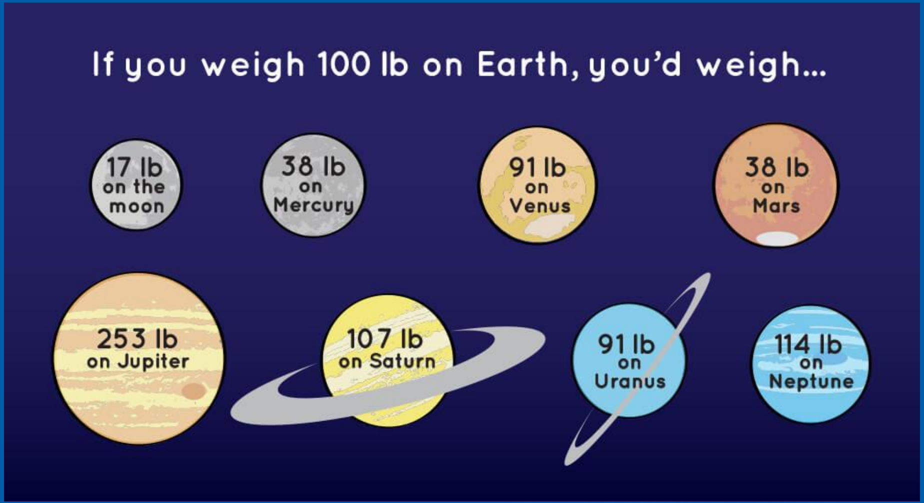
- **What's the difference velocity and acceleration?**
  - Acceleration is the rate of change of velocity!
  - Velocity is the rate of change of position!
  - They are independent! We can have 0 velocity with high acceleration and vice versa

Some Useful Concepts

Mass vs Weight

- Mass is the amount of stuff! It is the same everywhere
- Weight is the number on the scale as it is sandwiched between a planet and an object
- Weight is a force!



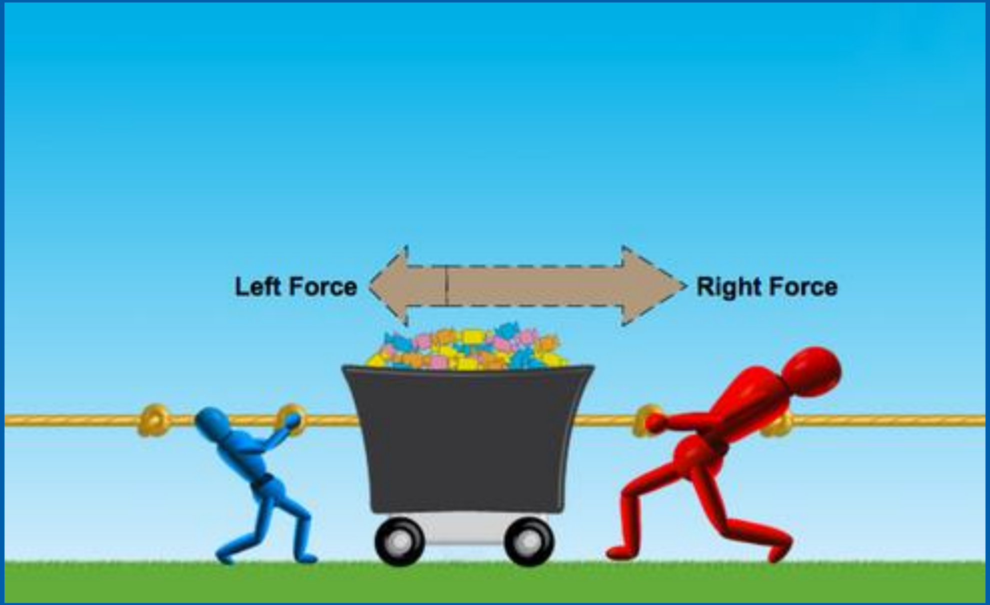


Key Concepts

A force is a push or a pull.

Net Force

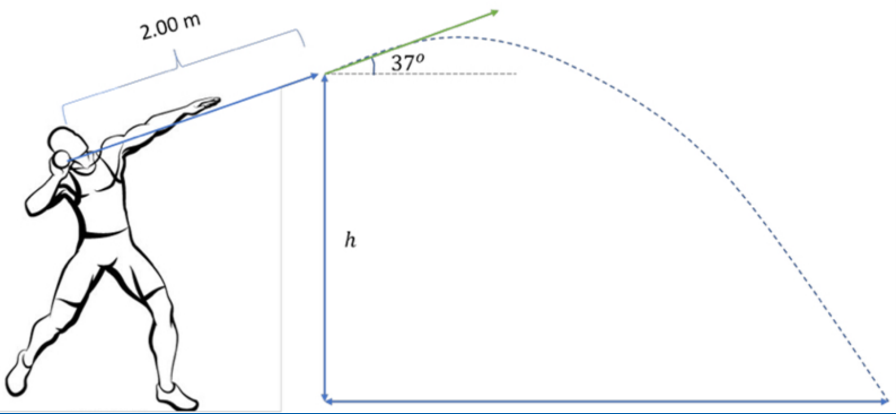
- Forces add like vectors!
- Opposite Pushes cancel out!



Review Newton's Laws of Motion

1. Newton's First Law of Motion **(Inertia)**	An object at rest remains at rest, and an object in motion remains in motion at constant speed and in a straight line unless acted on by an unbalanced force.
2. Newton's Second Law of Motion **(Force)**	The acceleration of an object depends on the mass of the object and the amount of force applied.
3. Newton's Third Law of Motion **(Action & Reaction)**	Whenever one object exerts a force on another object, the second object exerts an equal and opposite force on the first.

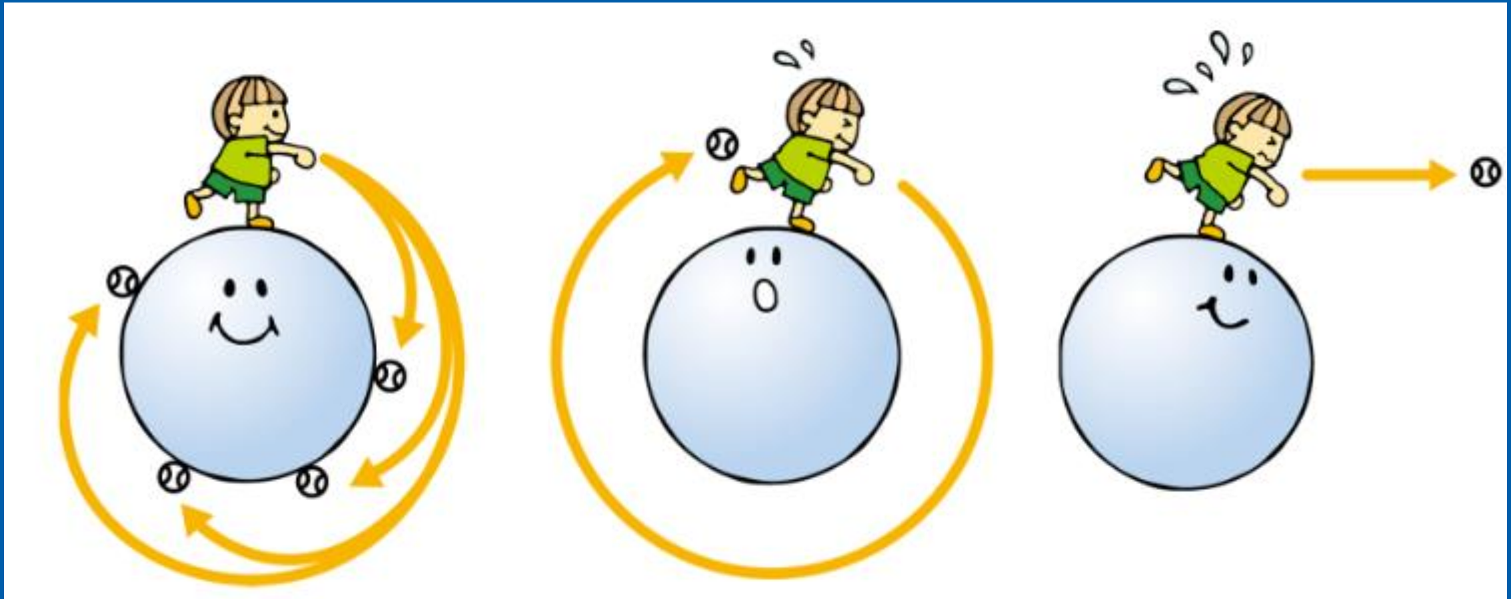
Motion and Gravity



Maddie Wesche wins bronze for New Zealand in the shot put at the 2025 Tokyo world TF Championships

Why Doesn’t the Moon fall down to Earth?

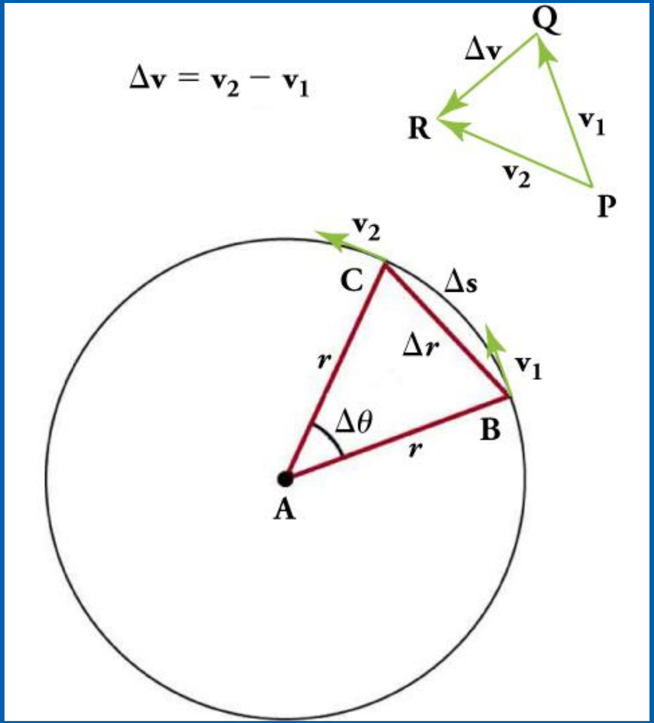
Why Doesn't the Moon fall down to Earth?



Answer: the Velocity has to be just right! Or in the right range at least.

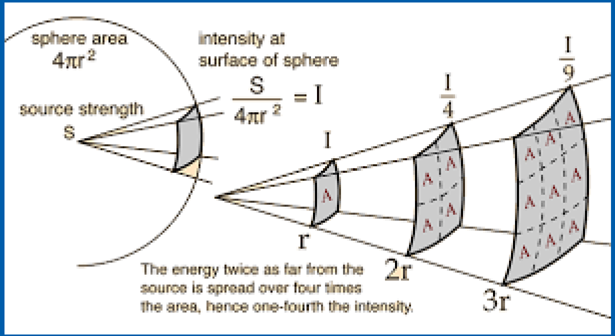
Why Doesn't the Moon fall down to Earth?

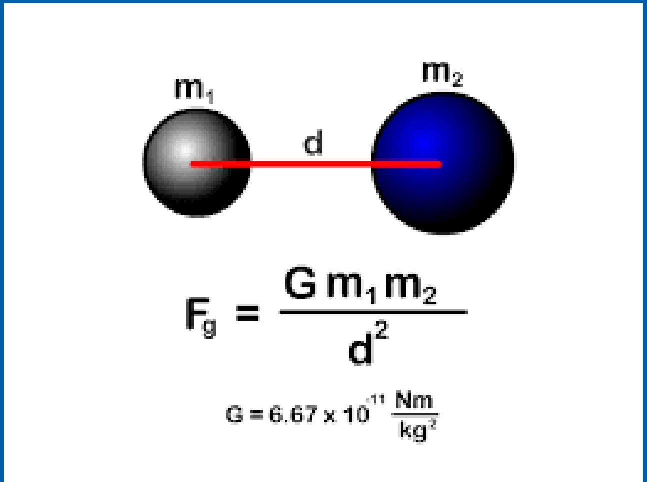
- The directions of the velocity of an object at two different points are shown, and the change in velocity Δv
- Δv is seen to point approximately toward the center of curvature (see small inset). For an extremely small value of Δs
- Δv points exactly toward the center of the circle (but this is hard to draw).
- ac=Δv/Δt, the acceleration is also toward the center, so ac is called centripetal acceleration.

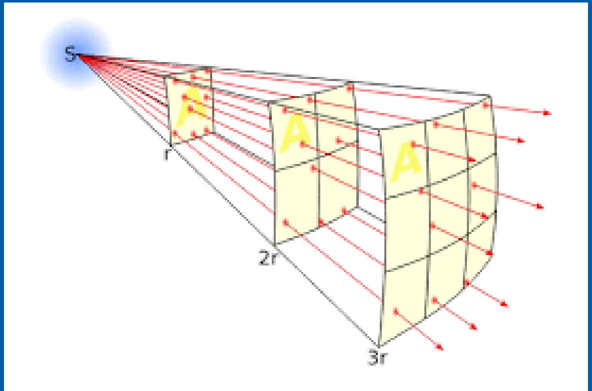


Gravitational Force

- Gravitational force gets weaker the further apart objects are.



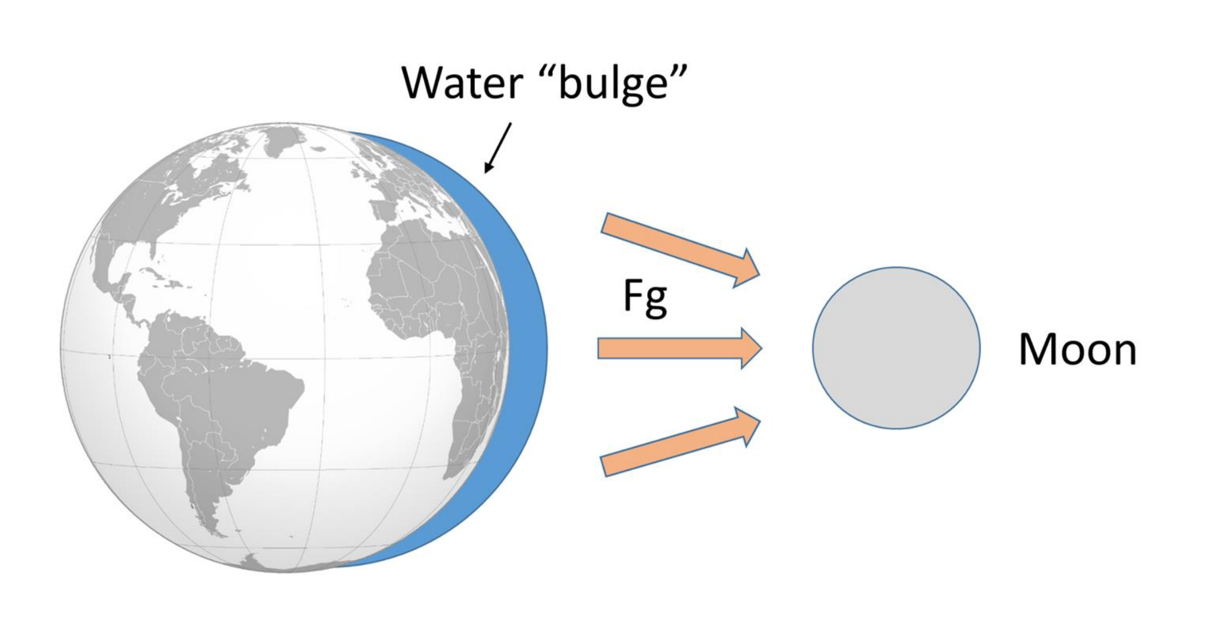




Some Useful Concepts

Tidal Forces

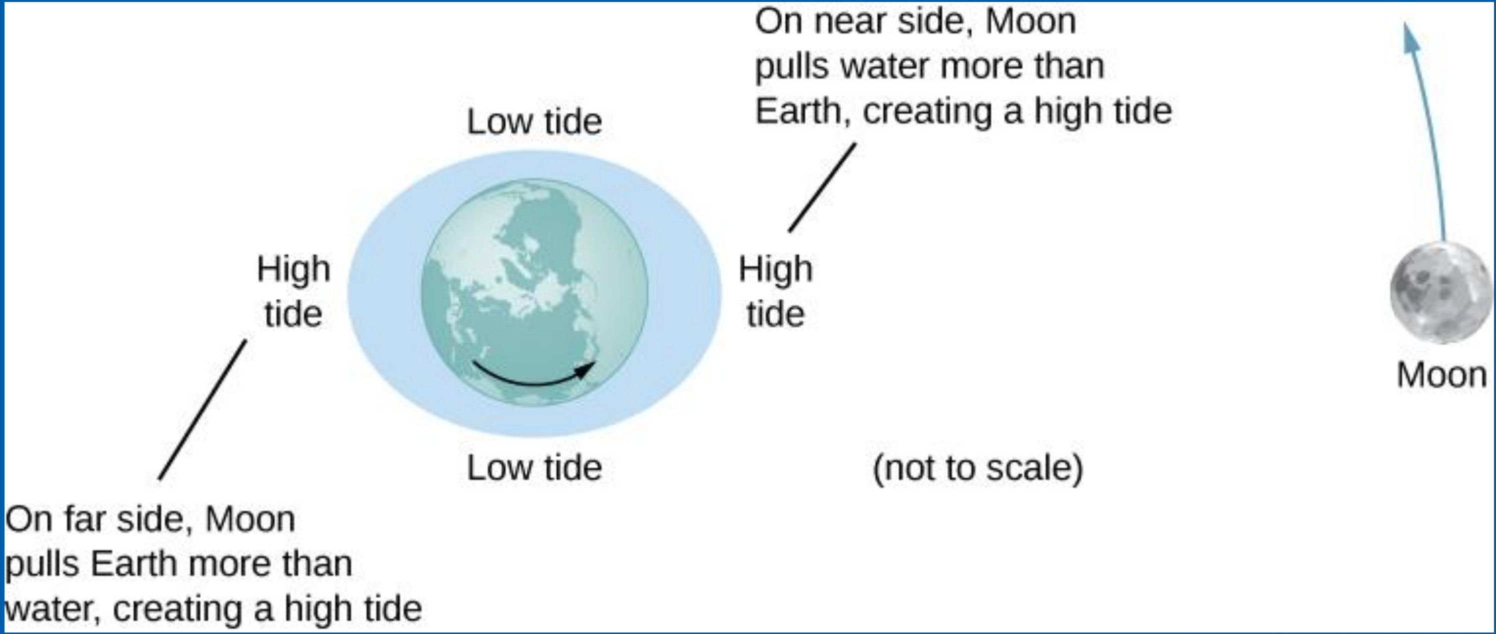
- Tides are due to the moon
- What's wrong with this picture?



Some Useful Concepts

Tidal Forces

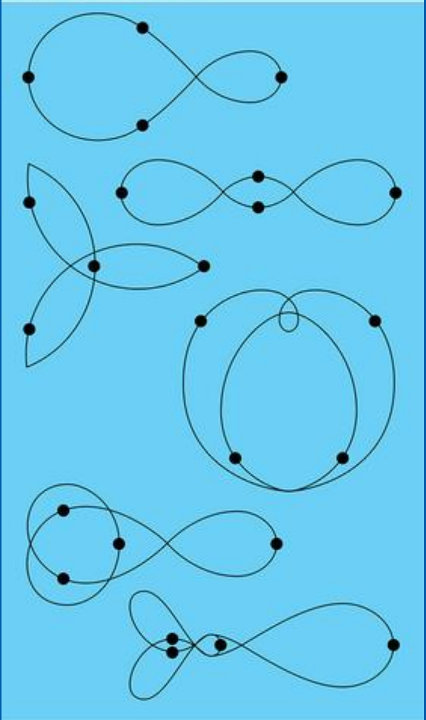
- High tide and low tide twice a day



Complex Gravitational Interactions

Gravity pulls everything to everything!

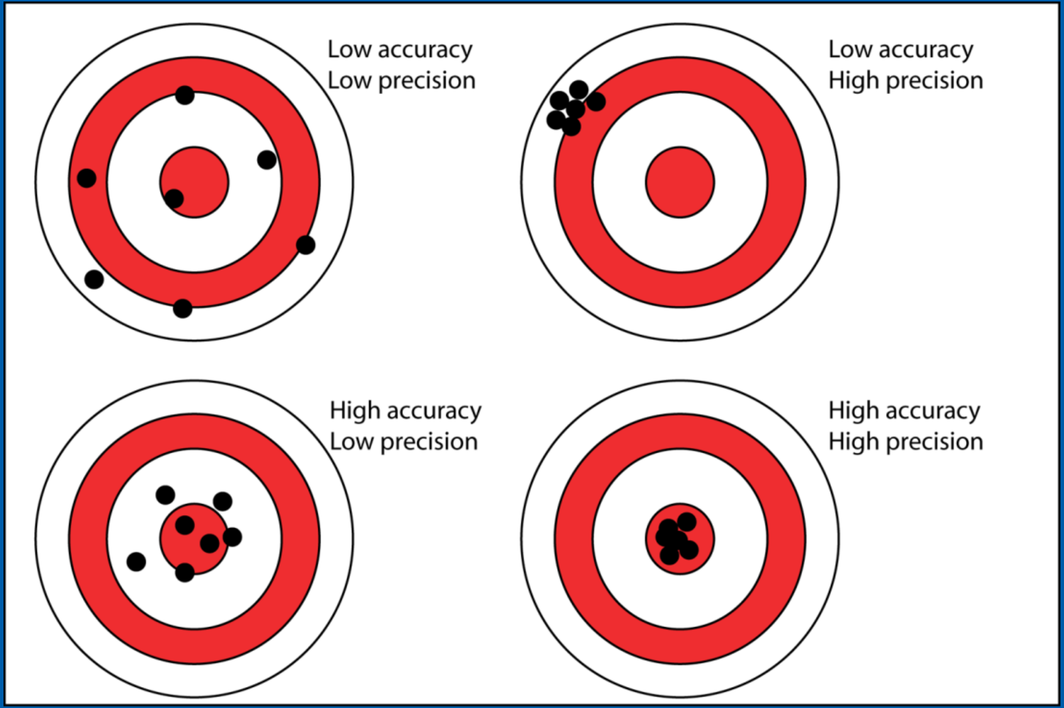
Figure 8: New Orbits



Some Useful Concepts

Precision vs Accuracy

- Precision is limited by your instruments
- Accuracy is determined by how your instruments are applied



Reminders

1. **Complete the Tutorial first!**
    - This is best done now in class where I am available to assist
1. **Attempt the essay questions last!**
    - These should be based on what you have covered in the tutorial.
1. **Previous week's lab report due at the beginning of the next lab!**
    - Final grades are based on best 6 online labs plus each of the 4 in-class labs
1. **Grades will be posted to Brightspace**
    - Check now that you have access to our course!
    - Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!
1. **Office hours**
    - Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!